

FINAL TARGETED SCIENTIFIC AUDIT & VALIDATION REPORT

Groundwater AI Heavy-Metal Screening System (North Bengal Aquifer Framework)

FINAL VALIDATION STATUS

- Statistical Validation:** PASS WITH LIMITATIONS (Small sample size $N = 40$, limited quantitative predictive power for Ni)
- Model Consistency:** PASS (Preserved exact binary `HuberRegressor` & `LinearRegression` pipelines)
- Preprocessing:** PASS (Integrated `Pipeline` with `SimpleImputer` & `StandardScaler`)
- Domain Validation:** PASS (2-Layer physical limits + range-based model domain check)
- Conformal Implementation:** PASS WITH LIMITATIONS (90% nominal out-of-fold conformal interval; observed coverage in evaluation artifact = 90.0%)
- Threshold Logic:** PASS (Conservative 5-state logic; zero "SAFE" terminology)
- Report Consistency:** PASS (10-section HTML report with unified recommendation block)
- Reproducibility:** PASS (Deterministic inference across 12 automated test cases)

OVERALL SCIENTIFIC AUDIT: PASS WITH LIMITATIONS

1. TARGETED ISSUE AUDIT SUMMARY TABLE (ISSUE 8 REQUIRED FORMAT)

Issue	Current State	Actual Verified State	Severity	Action
Training N	39	40 (Model estimators fit on $N = 40$ via <code>SimpleImputer(strategy='median')</code> ; complete-case domain profiling evaluated $N = 39$ due to 1 missing <code>WELL_DEPTH</code> in sample <code>S98_01798</code>)	LOW	Preserved models unchanged ($N = 40$). Explicitly documented complete-case vs imputed sample trace in manifest & reports.
Conformal wording	90% empirical coverage	90% nominal out-of-fold conformal prediction interval; observed coverage in available evaluation artifact = 90.0% (36/40 samples covered)	MEDIUM	Updated all UI cards, HTML reports, and audit logs to distinguish nominal interval designation from observed evaluation coverage.
OOF R^2 reporting	Difference 0.000	Primary (5×5 Nested CV): $Ni=0.2240$, $Cd=0.7061$ Secondary (Single-pass OOF): $Ni=0.2271$, $Cd=0.7158$ Actual Diff: $Ni = +0.0031$, $Cd = +0.0097$	MEDIUM	Reported Primary and Secondary metrics separately with actual non-zero differences. Stopped rounding differences to zero.
Scientific validity	PASS	PASS WITH LIMITATIONS (due to small $N = 40$, low $Ni R^2$, empirical association nature, and preliminary screening scope)	HIGH	Replaced blanket "PASS" with granular component scores and explicit scientific limitation statements.

2. DETAILED BREAKDOWN OF THE FOUR CORE AUDIT ISSUES

ISSUE 1 — TRAINING N TRACE & ROW-LEVEL DATA FLOW

[illegible]

- **Target | Original N | Final Training N | Excluded N | Excluded Sample(s) | Reason | Justified?:**
- **Nickel (Ni):** Original N = 40 | Final Training N = **40** | Excluded N = 0 in model fitting (1 in complete-case profiling) | Sample S98_01798 | Missing WELL_DEPTH (NaN) imputed via median in sklearn Pipeline | **YES**. Complete-case profiling evaluated N = 39 complete rows, while the ML model estimator utilized all N = 40 samples.
- **Cadmium (Cd):** Original N = 40 | Final Training N = **40** | Excluded N = 0 in model fitting (1 in complete-case profiling) | Sample S98_01798 | Missing WELL_DEPTH (NaN) imputed via median in sklearn Pipeline | **YES**. Identical dataset handling applied to Cd.

ISSUE 2 — CONFORMAL COVERAGE & TERMINOLOGY AUDIT

Inspection of CONFORMAL_NI.csv and CONFORMAL_CD.csv : - **Total evaluation observations:** 40 samples (S98_01785 through S98_01834). - **Covered count (is_covered == True):** 36 out of 40 samples for both Ni and Cd. - **Observed empirical coverage rate:** $36/40 = 90.0\%$.

Terminology Standards Enforced:

1. **Nominal Conformal Level:** 90% (the mathematical quantile target $q_{0.90}$ selected for interval width calculation).
2. **Observed Empirical Coverage:** 90.0% (36/40 samples covered in the evaluation artifact).
3. **Standard Wording:** "90% nominal CV+ conformal prediction interval; observed coverage in the available evaluation artifact = 90.0% (36/40)."
4. **Strictly Prohibited Claims:** Neither UI nor report claims "90% probability that true value lies inside interval" nor "guaranteed future coverage".

ISSUE 3 — OOF R² REPORTING & ACTUAL NON-ZERO DIFFERENCES

Target Metal	Primary Research Metric (5×5 Repeated Nested CV Mean)	Secondary Diagnostic Metric (Single-Pass OOF Concatenation)	Actual Difference (Secondary - Primary)	Primary vs Secondary Definition
Nickel (Ni)	$R^2_{\text{Repeated-OOF}} = 0.2240$ (RMSE = 1.7707 $\mu\text{g/L}$)	$R^2_{\text{Single-OOF}} = 0.2271$ (RMSE = 1.7675 $\mu\text{g/L}$)	+0.0031	Primary: Mean score across 25 fold evaluations in 5×5 nested CV (TABLE_4). Secondary: Global R^2 on concatenated 5-fold OOF vector (MODEL_PERFORMANCE_SUMMARY).
Cadmium (Cd)	$R^2_{\text{Repeated-OOF}} = 0.7061$ (RMSE = 0.0658 $\mu\text{g/L}$)	$R^2_{\text{Single-OOF}} = 0.7158$ (RMSE = 0.0648 $\mu\text{g/L}$)	+0.0097	Primary: Mean score across 25 fold evaluations in 5×5 nested CV (TABLE_4). Secondary: Global R^2 on concatenated 5-fold OOF vector (MODEL_PERFORMANCE_SUMMARY).

Note: Previous audit displayed "Difference = 0.000" due to over-rounding. The true non-zero differences (+0.0031 for Ni and +0.0097 for Cd) are now explicitly reported.

ISSUE 4 — AUDIT GRANULARITY & SCIENTIFIC LIMITATIONS

The blanket label "Scientific Validity: PASS" has been permanently replaced with **PASS WITH LIMITATIONS**.

Explicit System Limitations Documented in All Reports:

- Small Sample Size:** Trained on $N = 40$ regional groundwater wells in North Bengal.
- Limited Ni Predictive Power:** Nickel model (HuberRegressor) exhibits limited quantitative predictive power ($R^2_{\text{Repeated-OOF}} = 0.2240$).
- Observational Data Scope:** Predictors represent empirical statistical associations, not established hydrochemical causation.
- Preliminary Screening Scope:** Outputs represent model-based preliminary screening tools, not regulatory compliance certificates or laboratory analytical confirmations.
- Domain Restrictiveness:** Validated applicability is strictly restricted to the empirical training bounds (pH 6.87–7.32, TDS 44.5–593.4 mg/L, $\text{NO}_3\text{-N}$ 0.10–12.50 mg/L, Depth 9.0–61.0 m).

3. 12-CASE AUTOMATED TEST SUITE MATRIX (100% PASSED)

Test ID	Test Scenario	Input Vector (pH, TDS, NO3, Depth)	Domain State	Ni Status Code	Cd Status Code	Test Result
1	Normal In-Domain Sample	(7.1, 220, 1.2, 25)	IN_DOMAIN	BELOW_THRESHOLD	BELOW_THRESHOLD	PASS
2	Low pH (Out-of-Domain)	(6.5, 220, 1.2, 25)	OUT_OF_DOMAIN	OUT_OF_DOMAIN	OUT_OF_DOMAIN	PASS
3	High TDS (Out-of-Domain)	(7.1, 750, 1.2, 25)	OUT_OF_DOMAIN	OUT_OF_DOMAIN	OUT_OF_DOMAIN	PASS
4	High NO3-N (Out-of-Domain)	(7.1, 220, 15.0, 25)	OUT_OF_DOMAIN	OUT_OF_DOMAIN	OUT_OF_DOMAIN	PASS
5	Deep Well (Out-of-Domain)	(7.1, 220, 1.2, 80)	OUT_OF_DOMAIN	OUT_OF_DOMAIN	OUT_OF_DOMAIN	PASS
6	Negative TDS (Invalid)	(7.1, -50, 1.2, 25)	INVALID	Rejected	Rejected	PASS
7	Out-of-Bounds pH (Invalid)	(20.0, 220, 1.2, 25)	INVALID	Rejected	Rejected	PASS
8	Non-Numeric Input	('abc', 220, 1.2, 25)	INVALID	Rejected	Rejected	PASS
9	Conformal Data Missing	Simulated missing artifact	IN_DOMAIN	CONFIDENCE_UNAVAILABLE	CONFIDENCE_UNAVAILABLE	PASS
10	Interval Overlaps Threshold	High prediction near threshold	IN_DOMAIN	UNCERTAIN	UNCERTAIN	PASS
11	Interval Below Threshold	Normal sample	IN_DOMAIN	BELOW_THRESHOLD	BELOW_THRESHOLD	PASS
12	Potential Exceedance	High prediction > threshold	IN_DOMAIN	POTENTIAL_EXCEEDANCE	POTENTIAL_EXCEEDANCE	PASS

4. COMPUTATIONAL MODEL MANIFEST

```
{
  "application": "Groundwater AI Heavy-Metal Screening Web Application",
  "version": "1.0.0-research-hardened",
  "framework": "North Bengal Aquifer Hydrochemistry Framework",
  "models": {
    "ni": {
      "target": "Nickel (Ni_num)",
      "unit": "µg/L",
      "model_type": "HuberRegressor (scikit-learn Pipeline)",
      "file_path": "app/models/ni_model/FINAL_MODEL_NI.joblib",
      "conformal_artifact": "app/models/ni_model/CONFORMAL_NI.csv",
      "feature_list": ["pH_proxy", "TDS_calc", "NO3-N_num", "WELL_DEPTH"],
      "training_n_imputed": 40,
      "training_n_complete_case": 39,
      "sample_exclusion_note": "Sample S98_01798 has 1 missing WELL_DEPTH; complete-case domain profiling",
      "oof_r2_primary_5x5_nested_cv": 0.2240,
      "oof_r2_secondary_single_pass": 0.2271,
      "oof_r2_actual_difference": 0.0031,
      "oof_rmse": 1.7707,
      "conformal_nominal_coverage": "90%",
      "conformal_observed_coverage": "90.0% (36/40 samples covered)",
      "screening_threshold": 20.0
    },
    "cd": {
      "target": "Cadmium (Cd_num)",
      "unit": "µg/L",
      "model_type": "LinearRegression (scikit-learn Pipeline)",
      "file_path": "app/models/cd_model/FINAL_MODEL_CD.joblib",
      "conformal_artifact": "app/models/cd_model/CONFORMAL_CD.csv",
      "feature_list": ["pH_proxy", "TDS_calc", "NO3-N_num", "WELL_DEPTH"],
      "training_n_imputed": 40,
      "training_n_complete_case": 39,
      "sample_exclusion_note": "Sample S98_01798 has 1 missing WELL_DEPTH; complete-case domain profiling",
      "oof_r2_primary_5x5_nested_cv": 0.7061,
      "oof_r2_secondary_single_pass": 0.7158,
      "oof_r2_actual_difference": 0.0097,
      "oof_rmse": 0.0658,
      "conformal_nominal_coverage": "90%",
      "conformal_observed_coverage": "90.0% (36/40 samples covered)",
      "screening_threshold": 3.0
    }
  }
}
```

5. RESEARCH-SAFE TERMINOLOGY & MODEL INTEGRITY CONFIRMATION

- **Zero ML Models Altered:** Confirmed that `FINAL_MODEL_NI.joblib` and `FINAL_MODEL_CD.joblib` were not retrained, refitted, or modified in any manner.
- **Forbidden Terms Verification:** Searching the entire application codebase (`grep`) confirms 0 occurrences of non-defensible terms ("safe", "guaranteed safe", "proves", "causes", "100% accurate", "regulatory compliance confirmed").